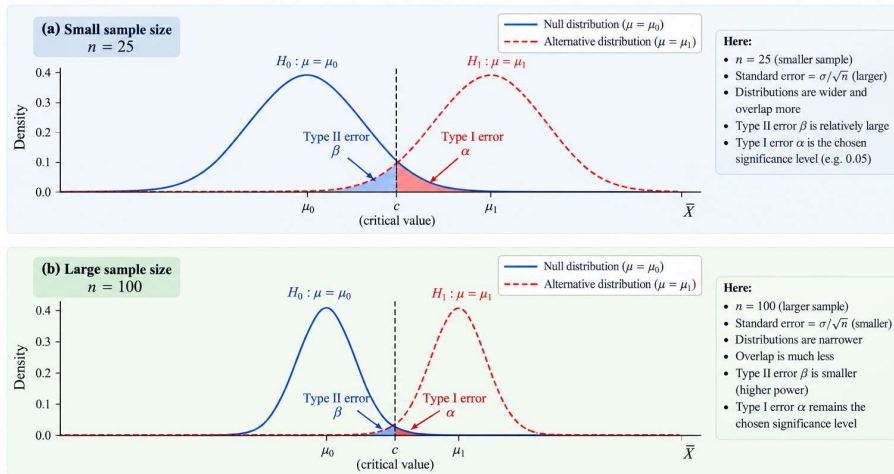


Non Parametric Tests

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Type I and Type II Errors and the Effect of Sample Size

Testing $H_0: \mu = \mu_0$ vs $H_1: \mu = \mu_1$ (one-sided, $\mu_1 > \mu_0$)



Key takeaway: Increasing n makes the sampling distributions narrower (standard error $\sigma/\sqrt{n}$ decreases), which reduces the Type II error β (increases power). Type I error α does not automatically decrease with n ; it is set by the chosen significance level.

→ Bayesian tests ? ⇔ Bayesian Credible Intervals

Introduction

Inferences based on z -statistics and t -statistics are **not always resistant** to outliers or violations of the assumptions

While large sample sizes can help to mitigate the problem at times, severe violations can make confidence levels for confidence intervals and p -values for hypothesis testing unreliable

Often, there are alternative methods that require fewer assumptions, if any

For testing, the name **nonparametric tests** is often used to refer to the family of such tests

While they are more widely applicable than tests that assume normality, equal variances, etc., they are also less efficient than the parametric methods

The less one assumes, the broader the applicability of the method, but also the less one can infer from the data

One assumption that stays though is that of independence!

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The Sign Test

The **sign test** is often used as a nonparametric alternative to the one-sample t -test

That is, to test the null hypothesis $\mu = \mu_0$ against a suitable alternative

For the sign test, we assume merely that the population sampled is **continuous** and **symmetrical**, **independent**

We assume that the population is continuous so that there is zero probability of getting a value equal to μ_0 (but ignore an observation that has a value equal to μ_0)

We do not even need the assumption of symmetry if we change the null hypothesis to $\tilde{\mu} = \tilde{\mu}_0$, where $\tilde{\mu}$ is the median

We replace each sample value exceeding μ_0 with a plus sign and each value less than μ_0 with a minus sign

Then we test the null hypothesis that the number of plus signs is a value of a random variable from $B(n, p = 1/2)$

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The Sign Test

With X being the number of positive signs, the distribution of X is binomial with parameters n and $p = 1/2$ under the null hypothesis that $\mu = \mu_0$

The mean of this distribution is $n/2$ and the variance is $n/4$

Large values of X suggest that $p > 1/2$ (corresponding to $\mu > \mu_0$); small values suggest that $p < 1/2$ (or $\mu < \mu_0$)

One can use the binomial distribution for exact p -value calculations

Or one can use a normal approximation if n is large; the z -ratio = $\frac{X - n/2}{\sqrt{n/4}}$

Note that there is no assumption about the population distribution

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EXAMPLE 16.1

The following are measurements of the breaking strength of a certain kind of 2-inch cotton ribbon in pounds:

163 165 160 189 161 171 158 151 169 162
163 139 172 165 148 166 172 163 187 173

Use the sign test to test the null hypothesis $\mu = 160$ against the alternative hypothesis $\mu > 160$ at the 0.05 level of significance.

Solution

1. $H_0: \mu = 160$
 $H_1: \mu > 160$
 $\alpha = 0.05$
2. Use the test statistic X , the observed number of plus signs.
3. Replacing each value exceeding 160 with a plus sign, replacing each value less than 160 with a minus sign, and discarding the one value that equals 160, we get
+ + + + - - + + + - + - + + + +
so $n = 19$ and $x = 15$. From Table 1 we find that $P(X \geq 15) = 0.0095$ for $\theta = \frac{1}{2}$.
4. Since the P -value, 0.0095, is less than 0.05, the null hypothesis must be rejected, and we conclude that the mean breaking strength of the given kind of ribbon exceeds 160 pounds.

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The Paired-Sample Sign Test

How can we modify this test for two-sample paired data?

For testing $H_0: \mu_1 = \mu_2$, we use the **paired-sample sign test** by computing the differences $d_i = x_i - y_i$ for $i = 1, \dots, n$

Replace positive differences by + and negative differences by -; use the sign test on these differences

EXAMPLE 16.3

To determine the effectiveness of a new traffic-control system, the numbers of accidents that occurred at 12 dangerous intersections during four weeks before and four weeks after the installation of the new system were observed, and the following data were obtained:

3 and 1, 5 and 2, 2 and 0, 3 and 2, 3 and 2, 3 and 0
0 and 2, 4 and 3, 1 and 3, 6 and 4, 4 and 1, 1 and 0

Use the paired-sample sign test at the 0.05 level of significance to test the null hypothesis that the new traffic-control system is only as effective as the old system. (The populations sampled are not continuous, but this does not matter as long as zero differences are discarded.)

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Solution to Example

Solution

1. $H_0: \mu_1 = \mu_2$
 $H_1: \mu_1 > \mu_2$
 $\alpha = 0.05$
2. Use the test statistic X , the observed number of plus signs.
3. Replacing each positive difference by a plus sign and each negative difference by a minus sign, we get

+ + + + + - + - + + +

so that $n = 12$ and $x = 10$. From Table I we find that $P(X \geq 10) = 0.0192$ for $\theta = \frac{1}{2}$.

4. Since the P -value, 0.0192, is less than 0.05, the null hypothesis must be rejected, and we conclude that the new traffic-control system is effective in reducing the number of accidents at dangerous intersections.

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The Wilcoxon Signed-Rank Test

The sign test, while easy to perform and requiring virtually no assumptions, only uses the signs and not the values of differences (differences with μ_0 for a one-sample test or paired differences for a two-sample test)

The Wilcoxon signed-rank test also makes use of the differences and uses therefore more information in the data. Here is how it works:

- Rank the data based on the absolute differences, averaging ranks when there are ties and ignoring zeros
- Compute the sum of the ranks for the positive differences, say T^+
- Then the sum of the ranks for the negative differences is $T^- = \frac{n(n+1)}{2} - T^+$
- Compute $T = \min(T^+, T^-)$

With $H_0: \mu = \mu_0$, rejection regions for level α tests are

- If $H_1: \mu \neq \mu_0$, then $\{T \leq T_{\alpha}\}$ (for T_{α} see Table X)
- If $H_1: \mu > \mu_0$, then $\{T^- \leq T_{2\alpha}\}$
- If $H_1: \mu < \mu_0$, then $\{T^+ \leq T_{2\alpha}\}$

Make changes as before for two-sample paired data

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The Wilcoxon Signed-Rank Test

The signed-rank test is especially good for heavy tailed distributions. It assumes a symmetric population distribution of differences

For $n \geq 15$, it is common to use a normal approximation

It can be shown that, under H_0 , the mean and standard deviation of T^+ are given by $n(n+1)/4$ and $\sqrt{n(n+1)(2n+1)/24}$, respectively

So, for $n \geq 15$,

$$z = \frac{T^+ - n(n+1)/4}{\sqrt{n(n+1)(2n+1)/24}}$$

can be used for testing in the usual way

An exact p -value can be computed using "wilcox.test" for paired data (in R)

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Example 16.4

EXAMPLE 16.4

The following are 15 measurements of the octane rating of a certain kind of gasoline: 97.5, 95.2, 97.3, 96.0, 96.8, 100.3, 97.4, 95.3, 93.2, 99.1, 96.1, 97.6, 98.2, 98.5, and 94.9. Use the signed-rank test at the 0.05 level of significance to test whether the mean octane rating of the given kind of gasoline is 98.5.

Solution

$$1. H_0: \mu = 98.5$$

$$H_1: \mu \neq 98.5$$

$$\alpha = 0.05$$

$$97.5 - 98.5$$

$$95.2 - 98.5$$

$$98.5 - 98.5 = 0$$

$$94.9 - 98.5$$

| x_i | D_i | $ D_i $ | R_i | $\text{sgn } R_i$ |
|-------|-------|---------|-------|-------------------|
| 98.2 | -0.3 | 0.3 | 1 | -1 |
| 99.1 | 0.6 | 0.6 | 2 | 2 |
| 97.6 | -0.9 | 0.9 | 3 | -3 |
| 97.5 | -1 | 1 | 4 | -4 |
| 97.4 | -1.1 | 1.1 | 5 | -5 |
| 97.3 | -1.2 | 1.2 | 6 | -6 |
| 96.8 | -1.7 | 1.7 | 7 | -7 |
| 100.3 | 1.8 | 1.8 | 8 | 8 |
| 96.1 | -2.4 | 2.4 | 9 | -9 |
| 96 | -2.5 | 2.5 | 10 | -10 |
| 95.3 | -3.2 | 3.2 | 11 | -11 |
| 95.2 | -3.3 | 3.3 | 12 | -12 |
| 94.9 | -3.6 | 3.6 | 13 | -13 |
| 93.2 | -5.3 | 5.3 | 14 | -14 |

Example 16.4

| Measurement | Difference | Rank |
|-------------|------------|------|
| 97.5 | -1.0 | 4 |
| 95.2 | -3.3 | 12 |
| 97.3 | -1.2 | 6 |
| 96.0 | -2.5 | 10 |
| 96.8 | -1.7 | 7 |
| 100.3 | 1.8 | 8 |
| 97.4 | -1.1 | 5 |
| 95.3 | -3.2 | 11 |
| 93.2 | -5.3 | 14 |
| 99.1 | 0.6 | 2 |
| 96.1 | -2.4 | 9 |
| 97.6 | -0.9 | 3 |
| 98.2 | 0.3 | 1 |
| 94.9 | -3.6 | 13 |

so that $T^- = 4 + 12 + 6 + 10 + 7 + 5 + 11 + 14 + 9 + 3 + 1 + 13 = 95$, $T^+ =$

$$T^- = 95, T^+ = \binom{15}{2} - T^- = 10, T = \min(T^-, T^+) = 10$$

Since $T < T_{.05} = 21$, we reject H_0 that the mean is 98.5; from Table X we see that the p -value is < 0.01

The `wilcox.test` function in R gives a p -value of 0.012

Example 16.4

While n is not very large here, let's also look at the large sample test

Using $n = 14$ (because one difference is equal to 0), we have that, under H_0 , the mean of T^+ is $n(n+1)/4 = 52.5$ and its variance is $n(n+1)(2n+1)/24 = 253.75$

$$\text{Hence, } z = (10 - 52.5) / \sqrt{253.75} = -2.668$$

The corresponding p -value is 0.00763, so that we would again reject H_0

Example 16.5

EXAMPLE 16.5

The following are the weights in pounds, before and after, of 16 persons who stayed on a certain reducing diet for four weeks:

| Before | After |
|--------|-------|
| 147.0 | 137.9 |
| 183.5 | 176.2 |
| 232.1 | 219.0 |
| 161.6 | 163.8 |
| 197.5 | 193.5 |
| 206.3 | 201.4 |
| 177.0 | 180.6 |
| 215.4 | 203.2 |
| 147.7 | 149.0 |
| 208.1 | 195.4 |
| 166.8 | 158.5 |
| 131.9 | 134.4 |
| 150.3 | 149.3 |
| 197.2 | 189.1 |
| 159.8 | 159.1 |
| 171.7 | 173.2 |

Use the signed-rank test to test at the 0.05 level of significance whether the weight-reducing diet is effective.

Solution

1. $H_0: \mu_1 = \mu_2$
 $H_1: \mu_1 > \mu_2$
 $\alpha = 0.05$

2. Reject the null hypothesis if $z \geq z_{0.05} = 1.645$, where

$$z = \frac{T^+ - \mu}{\sigma}$$

and μ and σ^2 are given by the formulas of Theorem 16.1.

3. The differences between the respective pairs are 9.1, 7.3, 13.1, -2.2, 4.0, 4.9, -3.6, 12.2, -1.3, 12.7, 8.3, -2.5, 1.0, 8.1, 0.7, and -1.5, and if their absolute values are ranked, we find that the positive differences occupy ranks 13, 10, 16, 8, 9, 14, 15, 12, 2, 11, and 1. Thus,

$$\begin{aligned} T^+ &= 13 + 10 + 16 + 8 + 9 + 14 + 15 + 12 + 2 + 11 + 1 \\ &= 111 \end{aligned}$$

$$\text{Since } \mu = \frac{16 \cdot 17}{4} = 68 \text{ and } \sigma^2 = \frac{16 \cdot 17 \cdot 33}{24} = 374, \text{ we get}$$

$$z = \frac{111 - 68}{\sqrt{374}} = 2.22$$

4. Since $z = 2.22$ exceeds $z_{0.05} = 1.645$, the null hypothesis must be rejected; we conclude that the diet is, indeed, effective in reducing weight.

Examples 16.4 and 16.5

The p -value of this approximate test is $P(Z > 2.22) = 0.0132$

Based on the (exact) Wilcoxon signed-rank test (see R), we obtain a p -value of 0.0124

The statistic V in R is identical to T^+ for Example 16.5

This is not the case for Example 16.4; I believe that this is caused by how ties are handled

If one had assumed normality in Examples 16.4 and 16.5 (are these reasonable assumptions?), then one could have used t -tests

For Example 16.4, this would have resulted in a p -value of 0.007; in Example 16.5 it would have been 0.004

The normality assumption for Example 16.4 seems to be just fine (see the plot in R); the assumption is a bit more suspect for Example 16.5

Rank-Sum Tests: The U Test

How about a nonparametric alternative to the two-sample t-test?

Such a test goes under different names: The **U test**, the **Wilcoxon test**, or the **Mann-Whitney test**

Without assuming normality of the two populations, we test the null hypothesis that we are sampling identical continuous populations against the alternative that the two populations are different

If we only consider a **shift in distributions**, then this is equivalent to testing equality of the medians; if we assume in addition symmetry, then it is a test for equal means

Arrange the values from both samples jointly (as if they were one sample) in an increasing order of magnitude and assign ranks

Tied observations get the mean of the ranks that they jointly occupy

If the populations are different (e.g., shifted), observations from one population would tend to get higher ranks than those from the other

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Rank-Sum Tests: The U Test

Calculate W_1 , the sum of the ranks corresponding to the first sample

Then $W_2 = \frac{(n_1+n_2)(n_1+n_2+1)}{2} - W_1$, where W_2 is the sum of the ranks for the second sample

Let $W = \min(W_1, W_2)$

In practice though, we consider slightly modified versions:

$$U_1 = W_1 - \frac{n_1(n_1+1)}{2}, \quad U_2 = W_2 - \frac{n_2(n_2+1)}{2},$$

and we let $U = \min(U_1, U_2)$

For symmetric, shifted distributions and with $H_0: \mu_1 = \mu_2$, rejection regions for level α tests are

- If $H_1: \mu_1 \neq \mu_2$, then $\{U \leq U_{\alpha}\}$ (for U_{α} see Table XI)
- If $H_1: \mu_1 > \mu_2$, then $\{U_2 \leq U_{2\alpha}\}$
- If $H_1: \mu_1 < \mu_2$, then $\{U_1 \leq U_{2\alpha}\}$

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Example 16.6

The study aims at comparing two types of emergency flares based on their burning time (rounded to the nearest tenth of a minute)

| | | | | | | | | | | |
|---------|------|------|------|------|------|------|------|------|------|------|
| Brand A | 14.9 | 11.3 | 13.2 | 16.6 | 17.0 | 14.1 | 15.4 | 13.0 | 16.9 | |
| Brand B | 15.2 | 19.8 | 14.7 | 18.3 | 16.2 | 21.2 | 18.9 | 12.2 | 15.3 | 19.4 |

Is the average burning time for Brand B flares longer than that for Brand A? Test at the 0.05 level of significance

$H_0: \mu_1 = \mu_2$ vs $H_1: \mu_1 < \mu_2$

$n_1 = 9, n_2 = 10$

$W_1 = 1 + 3 + 4 + 5 + 7 + 10 + 12 + 13 + 14 = 69$, so that

$U_1 = 69 - \frac{10}{2} = 24$

Based on Table XI, $U_1 = U_{0.1} = 24$, so that we reject H_0 at $\alpha = 0.05$; the data provide evidence that the burning time for flares of Brand B exceeds that for flares of Brand A

The `wilcox.test` function gives a p -value of 0.047 (with $W = 24$)

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Rank-Sum Tests: The U Test

If n_1 and n_2 are both greater than 8, it is reasonable to assume that U_1 and U_2 are values of random variables having approximately **normal distributions**. The following result provides the mean and variance of U_1 and U_2 under H_0

THEOREM 16.2. Under the assumptions required by the U test, U_1 and U_2 are values of random variables having the mean

$$\mu = \frac{n_1 n_2}{2}$$

and the variance

$$\sigma^2 = \frac{n_1 n_2 (n_1 + n_2 + 1)}{12}$$

For Example 16.6, $\mu = 45$, $\sigma^2 = 150$, so $z = \frac{U_1 - 45}{\sqrt{150}} = -1.715$, which corresponds to a p -value of 0.043. The **H test**, also called the **Kruskal-Wallis test**, is a generalization of the rank-sum test (U -test) to the case where we test the null hypothesis that k samples come from identical continuous populations.

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The Rank Correlation Test

Under normality we used r to test for correlation between two samples

One of the nonparametric measure of association is the **rank correlation coefficient**, also called **Spearman's rank correlation coefficient**, r_S .

DEFINITION 16.1. RANK CORRELATION COEFFICIENT. The rank correlation coefficient is given by

$$r_S = 1 - \frac{6 \sum_{i=1}^n d_i^2}{n(n^2 - 1)}$$

where d_i is the difference between the ranks assigned to x_i and y_i .

Tied observations get the mean of the ranks that they jointly occupy

If there are no ties in rank, r_S actually equals the correlation coefficient r calculated for the ranks

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The Rank Correlation Test

The null hypothesis of no correlation, indeed, is the same as the test of the null hypothesis that the x 's and y 's are randomly matched

For small values of n ($n \leq 10$), exact tests can be employed, but we will mostly deal with the large sample case

In the large sample case, the distribution of the random variable associate with r_S , that is of R_S , can be approximated closely with a normal distribution

THEOREM 16.4. Under the null hypothesis of no correlation, the mean and the variance of R_S are

$$E(R_S) = 0 \quad \text{and} \quad \text{var}(R_S) = \frac{n-1}{12}$$

Strictly speaking, the theorem applies when there are no ties, but it can be used unless the number of ties is large.

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